Classes & Objects (Foundation)

Class & Object

What is a Class?

- A **class** is a **blueprint**
- It defines **what an object will have (data)** and **what it can do (methods)**

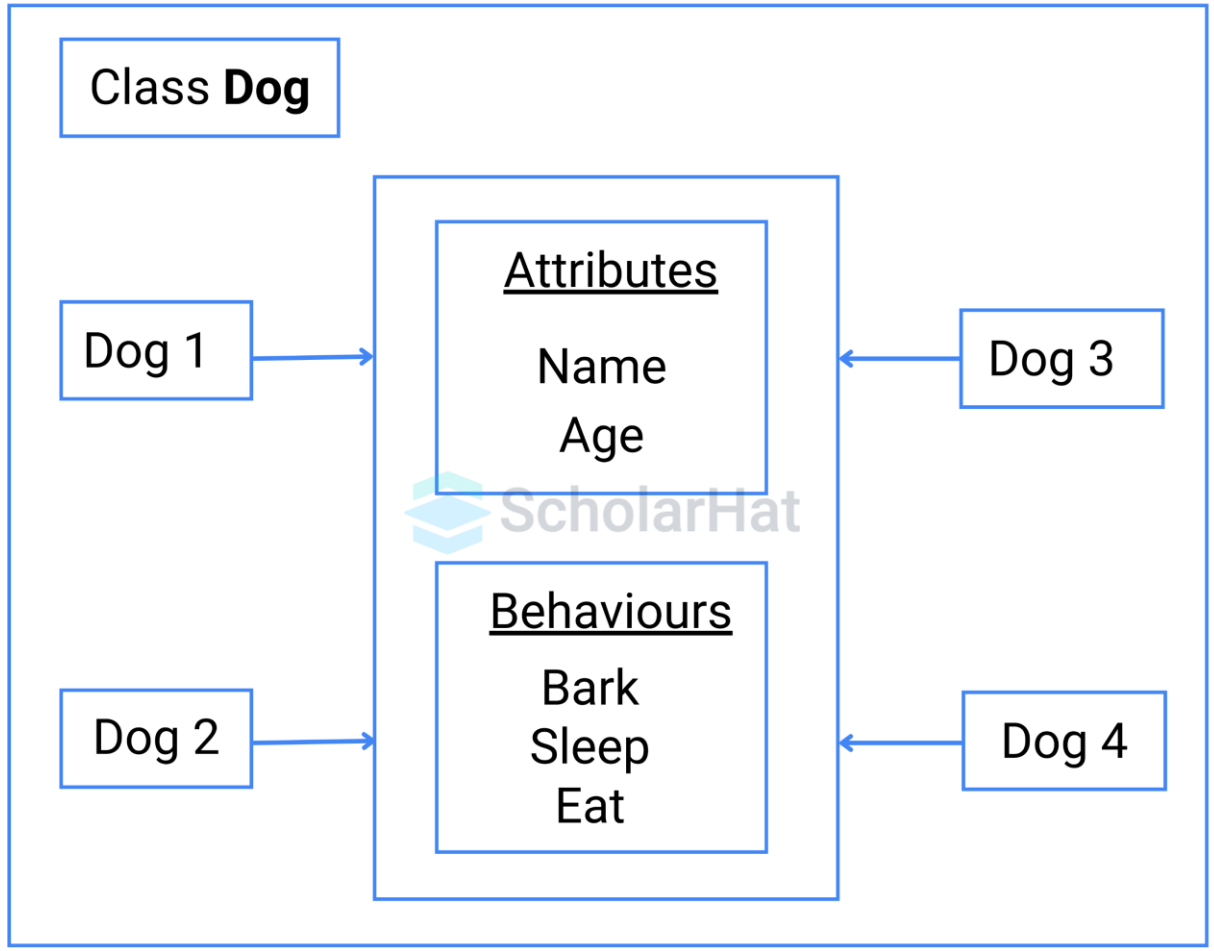
Real-World Example

- **Class** → Car
- **Objects** → BMW, Audi, Tesla

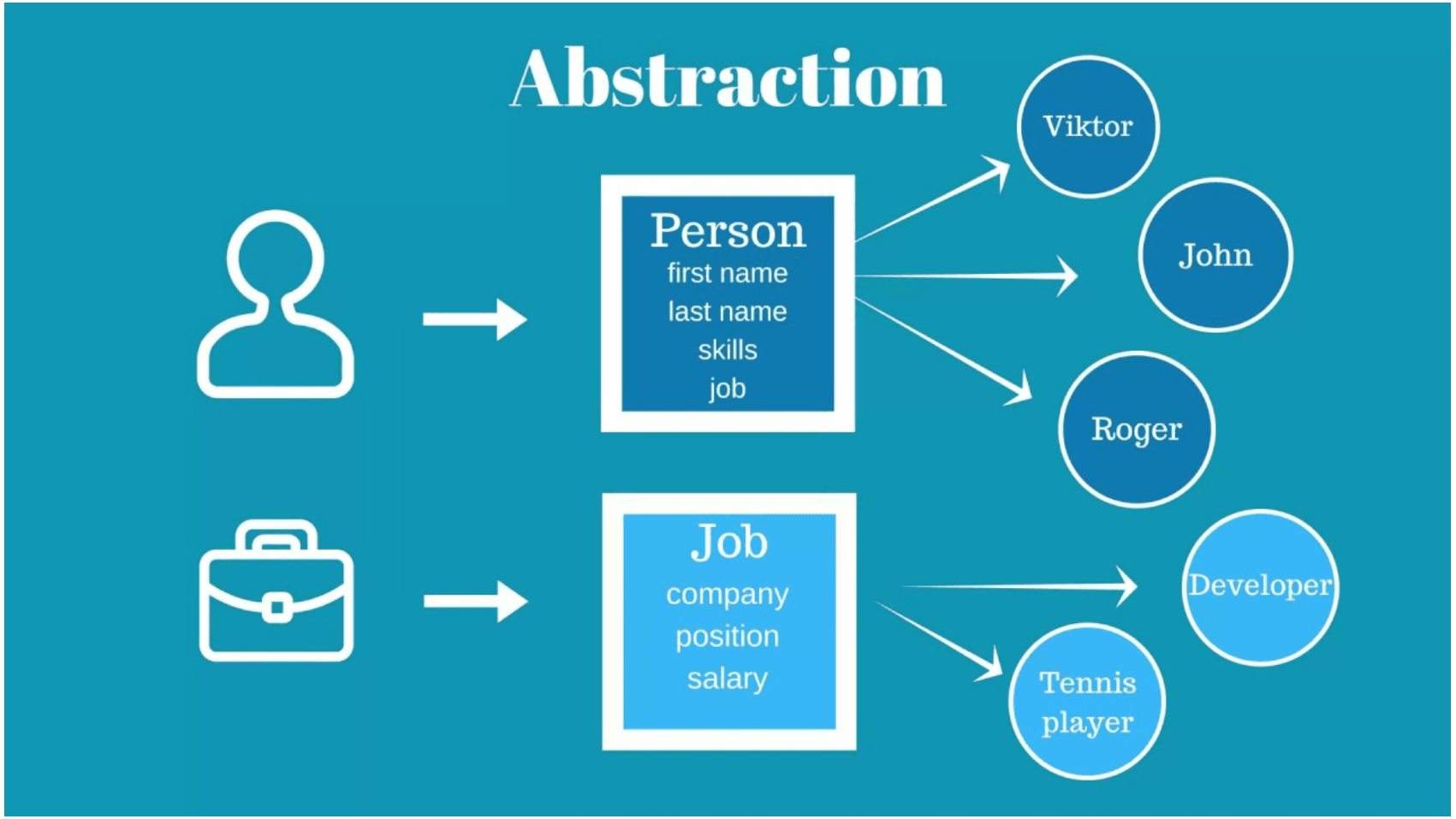
What is an Object?

- An **object** is a **real instance of a class**

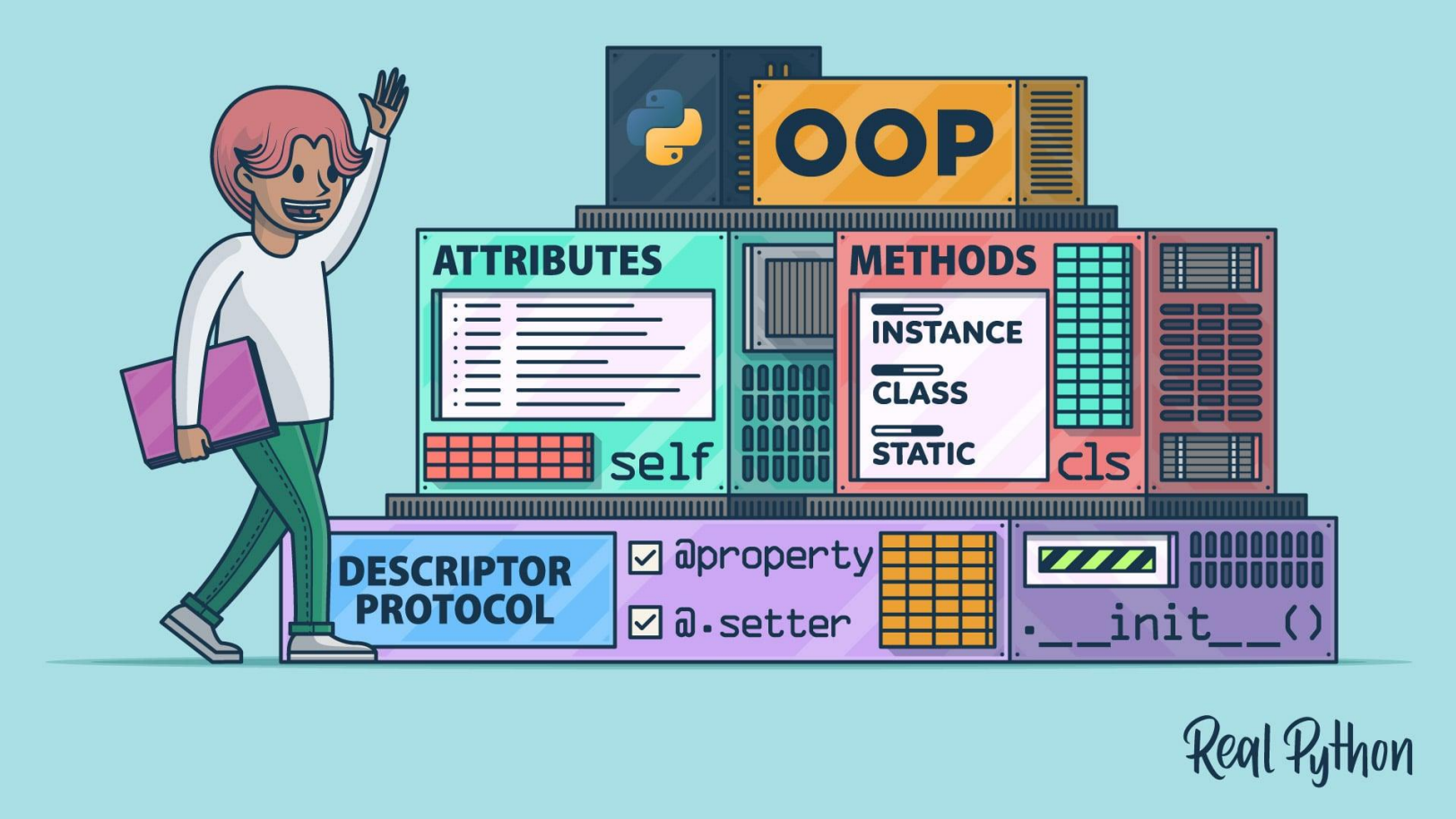
Class and Objects: Real-world Example



Class and Objects: Real-world Example



Class and Objects:



Simple Python Example

```python
class Student:
    def study(self):
        print("Student is studying")

# creating object
s1 = Student()
s1.study()
```

- Class Student --> blueprint
- s1 --> Object (real student)
- study() --> behaviour

Simple Python Example 2

```python
class Mobile:
    def call(self):
        print("Calling...")

    def sms(self):
        print("Sending SMS...")

m1 = Mobile()
m1.call()

m2 = Mobile()
m2.sms()
```

Question:

- Class Mobile --> ?
- m1 --> ?
- call() -->?
- m2 --> ?
- Sms() --> ?

Key Takeaways

- Class = blueprint
- Object = real thing
- Methods = actions
- Multiple objects can be created from one class